SATHIYANARAYANAN PALANI

Senior Software Engineer (Transitioning to AI Safety) · Interpretability & Civic Technology

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PROFESSIONAL SUMMARY

Software Engineer with **9+ years** building production systems and **2+ years** developing LLM-powered applications for civic impact.

- Implemented Anthropic's Sparse Autoencoder methodology for LLM interpretability research across **4 projects**
- **Founding engineer** at Ielektron for 4.5 years (one of the first 5 employees, through growth to ~300); later built telemetry infrastructure, ETL pipelines and web applications at **Intel**
- Published **6 PyPI packages** with **86k+ downloads**
- Built **FNMA**, a fact-checking platform (React, Python, Flutter, LLM APIs) against misinformation, WEF's #1 global risk (2024–25 reports)
- Eager to help mission-driven organizations (education, healthcare, nonprofits) deploy AI safely and effectively

KEY PROJECTS · LLM & MISSION-DRIVEN WORK

Cross-Model Generalization of Emotion-Vector Monitoring (Deployed Phase 0)

Oct 2026

Article: Cross-Model Generalization of Emotion-Vector Activation Monitoring

- Activation-based emotion monitoring (12 emotion directions plus neutral) tested across **7 models, 6 families, 1B–20B** (49 runs)
- Readings stayed stable under temperature and seed; all 7 models tracked nearly the same emotional arc of a scripted conversation
- Causal test: **stoic steering** compressed the emotion range by **63–83% in all 7 models**; an equal-size **sham** never reproduced it
- Built per-prompt median transition tables and raw sentence-level transient excerpts showing single-sentence emotion flips

Models: Gemma 3 (1B, 12B), Llama 3.2 1B, Qwen3 1.7B, Mistral 7B, GLM-4 9B, gpt-oss-20b

QBMP – Quoins, Bricks, Mortar & Pointing

Sep 2026

PyPI: QBMP · Repo

- Python library for **declarative synthetic dataset generation** via mathematical modelling of inputs, rules and weights
- **Sparse-to-dense** sampling in output space: quoin landmarks first, brick passes tile the gaps, mortar fills empty bins
- Solves non-monotonic rules; a coherence check re-derives every output; save() writes CSV/XLSX and an HTML datasheet

Tech: Python 3.11+, pandas, openpyxl, HTML/SVG/JS

Abliteration & Constitutional Classifiers++

Jun 2026 – Aug 2026

Article: Refusal-Direction Abliteration & Constitutional-Classifiers++ Pipeline · Repo

- Extracted the refusal direction from Falcon3-1B-Instruct via runtime activation ablation (no weight modification)
- Implemented Anthropic's two-stage Constitutional Classifiers++: **FastGate** activation probe + **ExchangeClassifier**
- 36-coordinate Cohen's-d signature matches the 38,912-coordinate baseline at the same **99.5% OOD accuracy** (0.09% of the information)
- Evaluated across **21 models** (0.6B–12B, 11 families, 27 h GPU); defence (91–99/100) generalises far better than attack (7–100/100)

Tech: PyTorch, Transformers, Ollama/gemma4, Plotly

LLM-Emotions – Interpretability Research

Mar 2026 – May 2026

Article: Sparse Autoencoder Analysis of Emotion Representations in Gemma 3 1B IT · Repo

- Trained an SAE (2304 latents, top-k = 32) on Gemma 3 1B IT with perfect feature disjointness (identity-matrix similarity)
- Recovered Russell's circumplex (valence and arousal axes) via PCA from raw activations; showed causal steering of emotional tone at inference

Tech: PyTorch, Transformers, scikit-learn, Matplotlib

Mechanistic Interpretability Projects

Jan 2026 – Mar 2026

Monosemanticity-MLP · Activation-Steering-et-Ablation · Feature-Ranking-et-Throttle-Testing

- Implemented Anthropic's *Towards Monosemanticity* on a toy MLP: data → MLP → activation harvesting → SAE → feature probing
- Monosemantic features with 60–66 active of 2304 latents (~3% sparsity); Sankey and heatmap views of input → neuron → feature → output circuits

Tech: PyTorch, pandas, Plotly

PyPI Open Source Packages

Jan 2022 – Apr 2022

PyPI: Palani-SN

- Published 6 Python packages incl. Py4Cli, LogExAn, ScEqAn (**86k+ lifetime downloads**)

Infrastructure as Code & Deployment

Sep 2025

Code2Cloud-Terraform-Helm

- Built end-to-end IaC pipeline: Terraform → Kubernetes → Helm charts for deploying web applications on GCP/GKE

Tech: Terraform, Helm, Kubernetes, GCP, Docker

FNMA – Civic Information Platform

May 2024 – Present

OnePager: River Banks, Proprioceptors & An Intangible Network · Kaggle

- Architected a full-stack misinformation detection platform: React website, Flutter app and Python backend with LLM-API fact-checking
- Designed **Adaptive Targeted Diffusion Control**, a self-regulating behavioral engine that reduces spam without manual moderation
- Built analytics with 7+ visualization types (heatmaps, trends, geo-spatial maps); designed for reproducible regional deployment

Tech: React, Python (FastAPI), ELK Stack, Claude/Gemini API, Flutter

LogExAnWebApp – Logical Expression Analysis

Mar 2022 – Oct 2022

Repo · Ansible deployment

- Solver that reverse-calculates logical expressions, finding the variable values that make a condition true or false
- FastAPI REST backend (DICT/JSON, DataFrame/Markdown) with a Flask UI; multi-environment deployment via Ansible

Tech: Python, FastAPI, Uvicorn, Flask, Ansible

PROFESSIONAL EXPERIENCE

Telemetry Software Data Engineer · Intel SRR, Bangalore

Nov 2023 – Present

- **ETL Pipeline:** Architected scalable data harvesting pipelines for bulk insertion to Elasticsearch, aggregating timeseries metrics from centralized TSDB cluster
- **Telemetry UI:** Built scalable web UI framework using MVC architecture and Dash, visualizing KPIs for distributed systems
- **EBDT (Event-Based Distributed Tracing):** Proposed and prototyped distributed tracing architecture inspired by OpenTelemetry for enhanced observability; submitted to ETTC 2025 (declined as out of scope)
- **QTOPP:** Built queued task processing system on process pools as headless CI/CD runner for scheduled statistical calculations
- **V&V POC:** Evaluated and prototyped infrastructure for AI-assisted verification and validation using pre-trained models

Semiconductor S/W Product Development Engineer · UST Global, Bangalore

May 2022 – Oct 2023

- Built Python automation test framework with GitHub Actions CI/CD and custom runners for telemetry testing
- Implemented Serial-Over-LAN using IPMI with monitoring threads for auto-retry during target reboot
- Collected performance KPIs from thermal/power subsystems via Redfish (DMTF spec) using OData queries over REST APIs
- Developed scripts to upload register values to InfluxDB using low-latency line protocol for real-time visualization

Senior Engineer (Founding Team) · Ielektron Technologies, Chennai

Oct 2018 – Dec 2021

One of the first 5 employees. Stayed as the company grew to ~300 over 4.5 years. Played a critical role in shaping initial phase of prototypes and guiding peers, operating as an early-stage key contributor during the formative phase of the startup.

Software Validation Engineer – MBRDI ADAS (as external)

Apr 2019 – Dec 2021

- Wrote Python test scripts for automated validation of ADAS measurements on HIL scenarios from DOORS requirements
- Automated HIL processes via SSH using paramiko and SCP
- Built custom framework for automated test case generation using pandas dataframes as pre-HIL execution task
- Performed runtime dynamic analysis of ADAS utility modules using ASTREE, unit/L2 testing with Google Test (C++)

Python Developer – SAFRAN Aerospace (as external)

Oct 2018 – Mar 2019

- Built multi-display broadcast system over local network using TCP/IP
- Created 3D scalar plot visualizations for sensor data using Plotly
- Developed event classification algorithm based on volumetric data normal distribution analysis

Embedded Software Developer (Founding Team) · Ielektron Technologies, Chennai

Jul 2017 – Sep 2018

- Improved embedded software unit and functional testing practices
- Built multi-branch CI/CD pipeline in Jenkins with GitHub app integration and Slack notifications
- Developed semi-automated hardware testing platform for Renesas microcontrollers using IronPython Renesas APIs

TECHNICAL SKILLS

Languages: Python, C/C++, Shell/Batch scripting
LLM & ML: PyTorch, Transformers, OpenCV, TensorFlow, Keras, scikit-learn, Sparse Autoencoders, activation steering, interpretability research, Claude Code, Copilot CLI, google-genai, agent architectures, evaluation frameworks, dspy, LangChain, LangGraph, Moshi-ASR, Moshi-TTS, delayed-streams-modelling
Web Development: FastAPI, Flask, React, Next.js, shadcn/ui
Data & Databases: pandas, polars, numpy, PostgreSQL, MySQL, SQLAlchemy, InfluxDB, Elasticsearch
Visualization: Matplotlib, Plotly, Dash
DevOps & Cloud: Docker, Kubernetes, Terraform, Helm, Jenkins, GitHub Actions, GCP (Cloud Run, Artifact Registry), AWS (EKS basics)
Protocols: TCP/IP, UDP, HTTP, MQTT, WebRTC, WebSockets, Redfish/OData
Testing: pytest, Google Test, unit/integration testing

EDUCATION

B.E. Electrical & Electronics Engineering · Anna University (AIHT)

2013 – 2017

First Class · Final Project: Automatic Attendance System using Face Recognition (OpenCV, Raspberry Pi)

PARTICIPATIONS & RECOGNITION

- European Test & Telemetry Conference 2025 – Paper Submission (2025)
- Google Gemini Long-Context Kaggle Competition Participant (2024)
- Civicus Nelson Mandela & Graca Machel Innovation Awards Participant (2024)
- OpenCV AI Scavenger Hunt – Week 4 Winner (2022)
- Hacktoberfest 2022 – Contributor
- 1st Prize – Inter-Department Working Model Development Competition during UG (2015)